AVA CHEN

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ACADEMIC POSITIONS

Stanford University **2025 – present**
Postdoctoral Scholar in Mechanical Engineering, Advisor: Allison Okamura
Wu Tsai Human Performance Fellow

EDUCATION

Columbia University **2019 – 2025**
Ph.D in Mechanical Engineering, Advisor: Matei Ciocarlie *New York, NY*
Thesis: *"Robotic Exoskeletons to Assist Volitional Motor Control of the Paretic Hand"*

Columbia University **2019 – 2021**
M.S. in Mechanical Engineering *New York, NY*

Massachusetts Institute of Technology (MIT) **2013 – 2017**
B.S. in Mechanical Engineering, Advisors: Hugh Herr, Arthur Petron *Cambridge, MA*
Thesis: *"Effectiveness of Active Cooling on Torque Performance for Prosthetic Applications"*

PUBLICATIONS

Peer-Reviewed Journal Articles

[∗ indicates equal contribution]

[J.5] J. Xu*, R. Wang*, S. Shang*, **A. Chen**, L. Winterbottom, L. Hsu, W. Chen, K. Ahmed, P. L. La Rotta, X. Zhu, D. M. Nilsen, J. Stein, and M. Ciocarlie, "ChatEMG: Synthetic Data Generation to Control a Robotic Hand Orthosis for Stroke." *IEEE Robotics and Automation Letters*, 10, 2, 907-914. (2025)
- Presented in *2025 IEEE Intl. Conference on Robotics and Automation (ICRA).*

[J.4] L. Winterbottom*, **A. Chen***, R. Mendonca, D.M. Nilsen, M. Ciocarlie, and J. Stein. "Clinician perceptions of a novel wearable robotic hand orthosis for post-stroke hemiparesis." *Disability and Rehabilitation*, 47, 6,1577–1586.(2024)

[J.3] J. Palacios*, A. Deli-Ivanov*, **A. Chen***, L. Winterbottom, D. M. Nilsen, J. Stein, and M. Ciocarlie, "Grasp Force Assistance via Throttle-based Wrist Angle Control on a Robotic Hand Orthosis for C6-C7 Spinal Cord Injury." *IEEE Transactions on Medical Robotics and Bionics*, 7, 1, 149-155. (2024)
- Presented in *2024 IEEE RAS/EMBS Intl. Conference on Biomedical Robotics and Biomechatronics (BioRob).*

[J.2] **A. Chen,** L. Winterbottom, S. Park, J. Xu, D. M. Nilsen, J. Stein, and M. Ciocarlie, "Thumb Stabilization and Assistance in a Robotic Hand Orthosis for Post-Stroke Hemiparesis." *IEEE Robotics and Automation Letters*, 7, 3, 8276-8282. (2022)
- Presented in *2022 IEEE RAS/EMBS Intl. Conference on Biomedical Robotics and Biomechatronics (BioRob).* **Finalist, BioRob2022 Best Paper Award**

[J.1] **A. Chen,** K. Kim, and P.S. Shamble. "Rapid mid-jump production of high-performance silk by jumping spiders." *Current Biology*, 31, R1422-R1423. (2021)

Peer-Reviewed Conference Papers

[C.8] **A. Chen**, M. C. Coram, C. du Pasquier, and A. M. Okamura. "Miniaturized Pneumatic Actuator Array for Multipoint Deep Pressure Tactile Stimulation." Work-in-Progress Paper in *2026 IEEE Haptics Symposium.*

[C.7] J. Xu*, **A. Chen*,** L. Winterbottom, J. Palacios, P. Chivukula, D. M. Nilsen, J. Stein, and M. Ciocarlie. "Reciprocal Learning of Intent Inferral with Augmented Visual Feedback for Stroke." In *2025 IEEE Intl. Conference on Rehabilitation Robotics (ICORR).*

[C.6] K. Lee, R. Wang, **A. Chen,** L. Winterbottom, H. M. C. Leung, L. M. DiSalvo, I. Xu, J. Xu, D. M. Nilsen, J. Stein, X. Zhou, and M. Ciocarlie. "Fabric Sensing of Intrinsic Hand Muscle Activity." In *2025 IEEE Intl. Conference on Rehabilitation Robotics (ICORR).*

[C.5] **A. Chen***, K. Lee*, L. Winterbottom, J. Xu, C. Lee, G. Munger, A. Deli-Ivanov, D. M. Nilsen, J. Stein, and M. Ciocarlie, "Volitional Control of the Paretic Hand Post-Stroke Increases Finger Stiffness and Resistance to Robot-Assisted Movement." In *2024 IEEE RAS/EMBS Intl. Conference on Biomedical Robotics and Biomechatronics (BioRob).*

[C.4] P. L. La Rotta*, J. Xu*, **A. Chen**, L. Winterbottom, W. Chen, D. M. Nilsen, J. Stein, and M. Ciocarlie, "Meta-Learning for Fast Adaptation in Intent Inferral on a Robotic Hand Orthosis for Stroke." In *2024 IEEE/RSJ Intl. Conference on Intelligent Robots and Systems (IROS).*

[C.3] **A. Chen,** L. Winterbottom, K. O'Reilly, S. Park, D. M. Nilsen, J. Stein, and M. Ciocarlie. "Design of Spiral-Cable Forearm Exoskeleton to Provide Supination Adjustment for Hemiparetic Stroke Subjects." In *2022 IEEE Intl. Conference on Rehabilitation Robotics (ICORR).*

[C.2] J. Xu, C. Meeker, **A. Chen,** L. Winterbottom, M. Fraser, S. Park, L. M. Weber, M. Miya, D. M. Nilsen, J. Stein, and M. Ciocarlie. "Adaptive Semi-Supervised Intent Inferral to Control a Powered Hand Orthosis for Stroke." In *2022 IEEE Intl. Conference on Robotics and Automation (ICRA).*

[C.1] T. Cervantes, W.E. Byun*, **A. Chen*,** K. Kim*, K. Nealon*, J. Connor, and A. Slocum. "A Device for Quantitative Analysis of the Thumb Ulnar Collateral Ligament." ASME. Frontiers in Biomedical Devices, *2018 ASME Design of Medical Devices Conference.*

Submitted for Publication

[S.3] C. du Pasquier*, A. Smith*, S. Huber, J. Phelps, I. A. Cohen, **A. Chen**, M. Kennedy, III, and A. M. Okamura. "Wearable Pneumatic Devices for Continuous, Closed-Loop, Bidirectional Tactile Interaction" (2026, under review)

[S.2] M. Polat*, S. Ahmed*, **A. Chen**, N. Gorantla, C. du Pasquier, A. Okamura, and A. G. Ramayya. "Motor response vigor tracks neurological function in patients with disorders of consciousness" (2026, under review)

[S.1] R. Wang*, X. Zhu*, **A. Chen**, J. Xu, L. Winterbottom, D. M. Nilsen, J. Stein, and M. Ciocarlie. "ReactEMG: Stable, Low-Latency Intent Detection from sEMG via Masked Modeling" (2025, under review)

Workshop and Symposium Contributions

[W.9] **A. Chen**, S. Ahmed, S. Choi, M. Markovich, M. Polat, E. Daly, A. Ramayya, and A. M. Okamura. "A Mechanical Stimulation Platform for Neurologial Assessment in Active Hospital Environments." In *2026 IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems (IROS) workshop: Out of the Lab: Personalization and Long-Term Use in Rehabilitation and Assistive Robotics.*

[W.8] L. Winterbottom, **A. Chen**, D. M. Nilsen, R. Mendonca, J. Xu, K. Lee, M. Ciocarlie, and J. Stein. "Motor learning techniques to enhance training with robotic hand orthoses for stroke survivors: challenges and opportunities." In *American Occupational Therapy Association (AOTA) INSPIRE 2025.*

[W.7] **A. Chen**, J. Xu, K. Lee, L. Winterbottom, D. M. Nilsen, J. Stein, and M. Ciocarlie. "Bidirectional Human-Robot Feedback and Physical Effects of Assisted Manipulation with a Robotic Hand Orthosis for Stroke." In *New England Manipulation Symposium (NEMS) 2024.*

[W.6] **A. Chen**. "Assisting Impaired Dexterity with User-Driven Robotic Hand Exoskeletons." In *Robotics: Science and Systems (RSS) workshop: RSS Pioneers 2024.*

[W.5] K. Lee, L. M. DiSalvo, I. Xu, **A. Chen**, X. Zhou, and M. Ciocarlie. "Fabric EMG Sensing for Robotic Orthosis Control." In *2024 IEEE Intl. Conference on Biomedical Robotics and Biomechatronics (BioRob) workshop: Building Responsive Body-Machine Interfaces with Biosignals and Robotic Exoskeletons.*

[W.4] L. Winterbottom, D. Nilsen, R. Mendonca, **A. Chen**, S. Lin, K. Carroll, J. Xu, M. Ciocarlie, and J. Stein. "Collaboration between Occupational Therapists, Engineers, and People with Neurological Conditions in the Development of Wearable Robotic Devices." In *American Occupational Therapy Association (AOTA) INSPIRE 2024.*

[W.3] J. Palacios*, A. Deli-Ivanov*, **A. Chen,** L. Winterbottom, D. M. Nilsen, J. Stein, and M. Ciocarlie. "Towards Tenodesis-Modulated Control of an Assistive Hand Exoskeleton for SCI." In *2023 IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems (IROS) workshop: Assistive Robotics for Citizens.*

[W.2] L. Winterbottom, K. Carroll, S. Lin, **A. Chen,** R. Mendonca, D. M. Nilsen, M. Ciocarlie, and J. Stein. "Stroke Survivors' Perspectives on the Design of a Novel Wearable Robotic Hand Brace." In *2022 Janet Falk-Kessler Distinguished Lectureship and Day of Scholarship.*

[W.1] L. Winterbottom, D. Nilsen, R. Mendonca, **A. Chen,** J. Xu, M. Ciocarlie, and J. Stein. "Perspectives of Individuals with C6-C7 Spinal Cord Injury on the Design of a Novel Robotic Hand Brace." In *2021 Janet Falk-Kessler Distinguished Lectureship and Day of Scholarship.*

Patents

[P.1] M. Ciocarlie, J. Stein, **A. Chen,** S. Park, D. M. Nilsen. "Robotic Hand Orthosis For Stroke", Application #: US 63/249,456

HONORS

Wu Tsai Human Performance Alliance Research Fellowship — **2026 – present**
NIH Ruth L. Kirschstein National Research Service Award (NRSA) F31 – NICHD — **2023 – 2025**
Robotics: Science & Systems (RSS) Pioneer — **2024**
Rising Star in ME 2022 at Stanford University — **2022**
Columbia University Presidential Distinguished Fellowship — **2019 – 2023**
Honorable Mention, MIT MechE deFlorez Design Competition — **2016**

INDEPENDENT FUNDING

On-field wearable haptic training to reduce knee injury risk during running cuts.
Stanford Wu Tsai Human Performance Research Fellowship 9/2026–8/2027 $94,867 + benefits **PI: Chen**

Impact of biofeedback and task-specific training with a robotic hand orthosis on voluntary muscle modulation for rehabilitation post-stroke. NIH F31 1F31HD111301 NICHD (NCMRR) 8/2023–1/2025 $72,587 **PI: Chen**

INVITED TALKS

Interpreting Human Action and Intent with Wearable Robots — **June 2026**
Stanford AI4ALL Guest Lecture

Enhancing Movement and Sensation with Haptic Garments — **Mar. 2026**
IEEE Haptics Symposium 2026, Cross-Cutting Challenge: Touch, Embodiment, and Multisensory Future

Assisting Human Autonomy with Wearable Robots — **Dec. 2025**
Duke University, Duke Robotics Seminar

Lessons in Dexterity From Observing Spiders, Humans, and Robots — **Dec. 2025**
Duke University, Robot Dexterity Lab (DexLab) Meeting

Robots to Assist and Observe Biological Autonomy, Behavior, and Interaction — **Dec. 2025**
UT Austin, MERGe Lab Meeting

Robotic Hand Exoskeletons to Assist and Rehabilitate Impaired Dexterity — **June 2024**
Harvard University, Harvard Biorobotics Lab Meeting

Robotic Hand Orthoses for Assistance and Rehabilitation After Stroke (Co-Speaker) — **Sept. 2023**
Global Perspectives on Medicine, Rehabilitation and Robotics Webinar Series

MyHand: a Wearable Hand Orthosis for Stroke (Co-Speaker) — **Oct. 2021**
IROS Workshop, Challenges and Opportunities of Human-Robot Symbiosis: from Wearable Robots to Neurorobotics

How Jumping Spiders Use Silk to Orient Themselves in Midair — **Oct. 2018**
Harvard University, Harvard Bauer Forum

How Jumping Spiders Jump — **Oct. 2018**
Broad Institute, CEE 35th Anniversary Celebration

TEACHING EXPERIENCE AND MENTORSHIP

University Courses
Teaching Assistant, Columbia MECE E4602 – Introduction to Robotics — **Fall 2020**
Lab Assistant, Harvard LS50 – Integrated Science — **Spring 2018, Spring 2019**

Pedagogical Training
Participant, Columbia Center for Teaching and Learning (CTL) Teaching Development Program — **2022 – 2025**

Talks on Teaching
Columbia CTL "Wowza!" CIRTL Discussion Series – Speaker, "Supporting Teaching as Scholarship" — **Mar. 2024**
Columbia CTLGrads Journal Club workshop – Speaker, "Effective Teaching Online, Real-Time" — **Oct. 2023**

Extracurricular
Research Mentor, Stanford University Summer Undergraduate Research Fellowship (SURF) — **2026**
Columbia Engineering Your PhD – Invited Panelist, "Insights from Experienced TAs" — **Aug. 2023, Aug. 2024**
Academic Mentor, Women in Science at Columbia (WISC) — **2020, 2021, 2023**
Research Mentor, Columbia University Engineering the Next Generation (ENG) — **Summer 2022**
Research Mentor and Teaching Assistant, Research Science Institute (RSI) — **Summer 2014**

RESEARCH STUDENTS SUPERVISED

Sunny Li, Stanford Ph.D Mechanical Engineering	**2025 – present**
Sarah Glomski, Stanford Ph.D Mechanical Engineering	**2025 – present**
Adarsh Ambati, Stanford Undergraduate	**2026 – present**
Sherry Choi, Stanford Undergraduate Visiting Intern [W.9]	**Summer 2026**
Jasmine Yang, Stanford SURF Fellow	**Summer 2026**
Cecilia Wu, Stanford M.S. Computer Science	**Fall 2025**
Andrew Chin, Columbia M.S. Robotics	**2024 – 2025**
Elise Yang, Columbia Undergraduate	**2024 – 2025**
Shiyao Marcus Lam, Columbia Undergraduate	**2024 – 2025**
Grace Munger, Columbia Undergraduate [C.5]	**2023 – 2025**
Matheu Campbell, Columbia Undergraduate	**2023 – 2024**
Akshay Venkatesan, Columbia M.S. Data Science	**2023 – 2024**
Runsheng Wang, Columbia M.A. Statistics [J.5, C.6, S.1, W.8]	**2023 – 2024**
Connor Lee, Columbia Undergraduate [C.5]	**2023 – 2024**
Alexandra Deli-Ivanov, Columbia Undergraduate [J.3, C.5, W.3]	**2022 – 2024**
Joaquin Palacios, Columbia Undergraduate and M.S. Robotics [J.3, C.7, W.3]	**2021 – 2024**
Pedro Leandro La Rotta, Columbia M.S. Robotics [C.4, J.5]	**2023**
Katherine O'Reilly, Columbia Undergraduate [C.3]	**2020 – 2023**
Carolyn David, Columbia M.S. Biomedical Engineering	**2022 – 2023**
Preethika Chivukula, Columbia M.S. Biomedical Engineering [C.7]	**2021 – 2022**
Ashley Reyes, Columbia ENG Student	**Summer 2022**
Brayan Ramos, Columbia ENG Student	**Summer 2022**
Ciara Little, Columbia Undergraduate	**2020 – 2021**
Katelyn G. Mitchell, Columbia Undergraduate	**2020 – 2021**
Frederick Horne, Harvard Undergraduate	**2019**
Rowen VonPlagenhoef, Harvard Undergraduate	**2019**
Eliot Burnes, Harvard Undergraduate	**2018 – 2019**
Henry Burnes, Harvard Undergraduate	**2018 – 2019**
Lincoln Sorscher, Harvard Undergraduate	**2018**
Cheng Lu, RSI Scholar	**Summer 2014**

SERVICE

University and Conference Service

Program Committee, RSS Pioneers 2025	**2025**
Workshop Co-Organizer, BioRob 2024	**2024**
"Building Responsive Body-Machine Interfaces with Biosignals and Robotic Exoskeletons"	
CIRTL Fellow, Columbia University Center for Teaching and Learning	**2023 – 2024**
Conference Volunteer, Robotics: Science and Systems (RSS)	**2022**

External Paper Reviewer

IEEE Robotics and Automation Letters (RA-L)	**2021, 2022, 2025, 2026**
International Journal of Robotics Research (IJRR)	**2026**
IEEE Transactions on Haptics (ToH)	**2025, 2026**
IEEE/RSJ Intl. Conference on Intelligent Robots and Systems (IROS)	**2024, 2026**
Eurohaptics	**2026**
IEEE RAS/EMBS Intl. Conference on Biomedical Robotics & Biomechatronics (BioRob)	**2022, 2024, 2026**
Robosoft	**2025**
Disability and Rehabilitation	**2025**
IEEE Intl. Conference on Rehabilitation Robotics (ICORR)	**2022, 2025**
IEEE Transactions on Robotics (TRO)	**2024**
IEEE Intl. Conference on Robotics and Automation (ICRA)	**2021, 2022, 2024**
IEEE Transactions on Medical Robotics and Bionics (T-MRB)	**2023**
Scientific Reports	**2022, 2023**
IEEE Intl. Conference on Robot and Human Interactive Communication (RO-MAN)	**2022, 2023**
IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)	**2020**

Science Volunteering and Outreach

Volunteer, Stanford Institutes of Medicine Summer Research Program	**2025**
Question Reviewer, U.S. Dept. of Energy National Science Bowl (NSB)	**2023, 2024**

Volunteer / Paper Reviewer, Research Science Institute (RSI) at MIT **2015, 2018, 2024 – 2026**
Columbia Engineering Achievers in Graduate Education (EngAGE) – Invited Panelist **Mar. 2024**
Columbia WISC STEM Field Exploration Fair – Invited Panelist, "Behind the Lab Scenes" **Apr. 2022**
Judge, Kentucky Science and Engineering Fair **2021**
Judge, MIT Mechanical Engineering Research Exhibition **2020**
Question Writer, USA Biolympiad (USABO) **2019**
Volunteer, Adaptive Climbing Group NY **2019**
Judge, Sweden Research Academy for Young Scientists (RAYS) **2015**

PREVIOUS POSITIONS

Harvard Dept. of Organismic & Evolutionary Biology, Shamble Lab **2017 – 2019**
Research Assistant with Paul Shamble
Studied jumping dynamics and mid-air silk production of jumping spiders (S. scenicus) [J.1]

Dephy, Inc. **Summer 2017, Fall 2018**
Mechanical Engineering Intern
Designed and built validation equipment for full-system stress testing

MIT Media Lab, Biomechatronics Group **2013 – 2017**
Undergraduate Researcher with Hugh Herr, Arthur Petron, and Matt Carney
Worked on FitSocket project for soft-tissue characterization and on active motor cooling

Apple Inc. **Summer 2016**
Product Design Validation Engineer Intern
Worked on design, usability, and validation for mechanical features in hardware products

Formlabs **Summer 2015**
Mechanical Engineering Intern
Electromechanical design for early versions of Form 2 and Form Cure products